describing the functioning of its methods. In section 2.6 we delve into the implementation of a d-way heap.

Section 2.7 and 2.8 will describe use cases where heaps and priority queues make a difference and allow speeding up applications or other algorithms.

Finally, the focus of section 2.9 will be understanding what's the optimal branching factor for a heap: although it can be considered optional, I suggest you at least try to read it through, to gain a deeper understanding of how heap works and why you might want to choose a ternary heap instead of a binary heap, or vice versa.

2.1 Structure of this chapter

Starting with the current chapter, we will embrace a schematic way to present our data structures.

Each chapter will be driven by a practical use case for its main topic, a real-world problem showing how the data structure is used in practice, but also step by step explaining how operations are performed; we also provide code samples showing how to use the algorithms we will focus on next.

So, first we are going to introduce a problem, one that is typically solved making use of the main topic of the chapter.

Then we present one or more ways to solve it: there might be several possible solutions to the same problem, and if that's the case, we explain when and why using a particular data structure is a good idea. At this point, usually, we still